

3.8 Dont care Conditions

A four-bit decimal code, for example, has six combinations which are not used. Any digital circuit using this code operates under the assumption that these unused combinations will never occur as long as the system is working properly. As a result, we don't care what the function output is to be for these combinations of the variables, because they are guaranteed never to occur. These don't care conditions can be used on a map to provide further simplification of the function.

To distinguish the don't care conditions from 1's and 0's, an X will be used. When choosing adjacent squares to simplify the function in the map, the X's may be assumed to be either 0 or 1, whichever gives the simplest expression. In addition, an X need not be used at all if it does not contribute to covering a larger area. In each case, the choice depends only on the simplification that can be achieved.

Example 3-12: Simplify the Boolean function:
 $F(w,x,y,z) = \Sigma(1,3,7,11,15)$
 and the don't care conditions: $d(w,x,y,z) = \Sigma(0,2,5)$

Map for $F(w,x,y,z)$ with don't care conditions marked as X:

w \ xz	00	01	11	10
00	X	1	1	X
01	X	X	1	
11			1	
10			1	

Map for $F = yz + w'z$:

w \ xz	00	01	11	10
00	X			X
01	0	X		0
11	0	0		0
10	0	0		0

X X

Map for $F' = z' + wy$:

w \ xz	00	01	11	10
00	X			X
01	0	X		0
11	0	0		0
10	0	0		0

$$\begin{aligned}
 F' &= z' + wy \\
 F &= (F')' = (z' + wy)' = (z')' \cdot (wy)' \quad (\text{Theorem 5(a)}) \\
 &= (z)' \cdot (w' + y') \quad (\text{Theorem 5(b)}) \\
 &= z \cdot (w' + y) \quad (\text{Theorem 3})
 \end{aligned}$$